Solution:

$xp^n = k \Rightarrow x = kp^{-n}$

$\frac{dx}{dp} = -nkp^{-n-1}$

Elasticity of demand : $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$

$= -\frac{p}{kp^{-n}}(-nkp^{-n-1}) = n$, which is a constant.

Example 6.8

For the given demand function $p = 40 - x$, find the output when $\eta_d = 1$

Solution:

$p = 40 - x \Rightarrow x = 40 - p$

$\frac{dx}{dp} = -1$

Elasticity of demand: $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp} = \frac{40 - x}{x}$

Given that $\eta_d = 1$

$\therefore \frac{40 - x}{x} = 1 \Rightarrow x = 20$

$\therefore$ output(x) $= 20$ units.

Example 6.9

Find the elasticity of demand in terms of x for the demand law $p = (a - bx)^{\frac{1}{2}}$. Also find the output$(x)$ when elasticity of demand is unity.

Solution:

$p = (a - bx)^{\frac{1}{2}}$.

Differentiating with respect to the price 'p', we get $1 = \frac{1}{2}(a - bx)^{\frac{-1}{2}}(-b) \cdot \frac{dx}{dp}$

$\therefore \frac{dx}{dp} = \frac{2(a - bx)^{\frac{1}{2}}}{-b}$

Elasticity of demand: $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$

$= -\frac{(a - bx)^{\frac{1}{2}}}{x} \cdot \frac{2(a - bx)^{\frac{1}{2}}}{-b} = \frac{2(a - bx)}{bx}$

When $\eta_d = 1$, $\frac{2(a - bx)}{bx} = 1$

$2(a - bx) = bx \Rightarrow$ output$(x) = \frac{2a}{3b}$ units.

Example 6.10

Verify the relationship of elasticity of demand, average revenue and marginal revenue for the demand law $p = 50 - 3x$.

Solution:

$p = 50 - 3x$

$\frac{dp}{dx} = -3 \Rightarrow \frac{dx}{dp} = -\frac{1}{3}$

Elasticity of demand: $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$

$= -\frac{50 - 3x}{x}\left(-\frac{1}{3}\right) = \frac{50 - 3x}{3x}$ ... (1)

Now, Revenue: $R = px$

$= (50 - 3x)x = 50x - 3x^2$

Average revenue: $AR = p = 50 - 3x$

Marginal revenue: $MR = \frac{dR}{dx} = 50 - 6x$

$\therefore \frac{AR}{AR - MR} = \frac{50 - 3x}{(50 - 3x) - (50 - 6x)}$

$= \frac{50 - 3x}{3x}$ ... (2)

From (1) and (2), we get

$\eta_d = \frac{AR}{AR - MR}$, Hence verified.

Example 6.11

Find the elasticity of supply for the supply law $x = \frac{p}{p + 5}$ when $p = 20$ and interpret your result.

Solution:

$x = \frac{p}{p + 5}$

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